

Week 3: Seller and Inventory Management System

Scenario:

The e-commerce company sells products supplied by multiple sellers. The system should maintain seller details, track products supplied by each seller, and monitor inventory levels to ensure product availability.

1. Create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(20), Seller_Name VARCHAR(30), Email VARCHAR(35), Phone NUMBER(10), Address VARCHAR(50));
```

Table created.

```
SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(10), Product_ID VARCHAR(10), Seller_ID VARCHAR(20), Stock_Quantity NUMBER, Stock_Status VARCHAR(20), Last_Update DATE);
```

Table created.

2. Establish Relationships

Seller and Product

One seller can supply multiple products.

One product can be supplied by one or multiple sellers.

Relationship:

Seller → Product is one to many

Product and Inventory

Each product has inventory information.

Inventory tracks available stock.

Relationship:

Product → Inventory is one to one

Primary Key Added for Both Tables

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);
```

Table altered.

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);
```

Table altered.

Foreign Key for Inventory Table

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_ID);
Table altered.
```

3. Constraints

Constraint to Make Unique Email ID

```
SQL> ALTER TABLE Seller ADD CONSTRAINT Un_mail UNIQUE(Email);
Table altered.
```

To Make Email NOT NULL

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;
Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);
Table altered.
```

4. Insert Values to Seller Table

```
SQL> INSERT INTO Seller VALUES('S00001','Vani Enterprises','mdvanienter@gmail.com',8765432109,'1, Anna Nagar, Chennai');
1 row created.

SQL> INSERT INTO Seller VALUES('S00002','SD & CO','seller0827@gmail.com',9876543012,'62, Thirumal Nagar, Thanjavur');
Enter value for co:
old 1: INSERT INTO Seller VALUES('S00002','SD & CO','seller0827@gmail.com',9876543012,'62, Thirumal Nagar, Thanjavur')
new 1: INSERT INTO Seller VALUES('S00002','SD ','seller0827@gmail.com',9876543012,'62, Thirumal Nagar, Thanjavur')
1 row created.

SQL> INSERT INTO Seller VALUES('S00003','AtoZ','atozmarket@gmail.com',7890654312,'90, George Town, Chennai');
1 row created.

SQL> INSERT INTO Seller VALUES('S00004','DSW Manufacturers','dsvmanuf142@gmail.com',7824003009,'23, Thillai Nagar, Trichy');
1 row created.
```

Display Seller Table

```
SQL> SELECT * FROM Seller;
```

SELLER_ID	SELLER_NAME	EMAIL	PHONE	ADDRESS
S00001	Vani Enterprises	mdvanienter@gmail.com	8765432109	1, Anna Nagar, Chennai
S00002	SD	seller0827@gmail.com	9876543012	62, Thirumal Nagar, Thanjavur
S00003	AtoZ	atozmarket@gmail.com	7890654312	90, George Town, Chennai
S00004	DSW Manufacturers	dsvmanuf142@gmail.com	7824003009	

5. Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('Inv0001','501','S00001',10,'Available',TO_DATE('2026-09-12','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv0002','502','S00002',15,'Available',TO_DATE('2026-09-13','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv0003','503','S00003',25,'Available',TO_DATE('2026-09-15','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv0004','504','S00004',30,'Available',TO_DATE('2026-09-16','YYYY-MM-DD'));
1 row created.
```

Display Inventory Table

```
SQL> SELECT * FROM inventory;
```

INVENTORY_	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS
Inv0001 12-SEP-26	501	S00001	10	Available
Inv0002 13-SEP-26	502	S00002	15	Available
Inv0003 15-SEP-26	503	S00003	25	Available
Inv0004 16-SEP-26	504	S00004	30	Available

Report 1: Seller-wise Product Report

Display:

- Seller name
- Number of products supplied
- Available stock

```
SQL> SELECT s.Seller_Name, COUNT(i.Product_ID) AS Number_of_Products, SUM(i.Stock_Quantity) AS Available_Stock FROM Seller s JOIN Inventory i ON s.Seller_ID = i.Seller_ID WHERE i.Stock_Status = 'Available' GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
SD	1	15
DSW Manufacturers	1	30
AtoZ	1	25
Vani Enterprises	1	10

Report 2: Stock Availability Report

Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status FROM Product p JOIN Inventory i ON p.Product_ID = i.Product_ID;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
Laptops	10	Available
Cameras	15	Available
Watches	25	Available
Bags	30	Available

Report 3: Inventory Analysis

1. Total Products Available

```
SQL> SELECT COUNT(*) AS Total_Available_Products FROM Inventory WHERE Stock_Status = 'Available';
```

TOTAL_AVAILABLE_PRODUCTS
4

2. Products Out of Stock

```
SQL> SELECT COUNT(*) AS OutOfStock_Products FROM Inventory WHERE Stock_Status = 'Out of Stock';
```

OUTOFSTOCK_PRODUCTS
0

3. Highest Stocked Products

```
SQL> SELECT p.Product_Name, i.Stock_Quantity FROM Product p JOIN Inventory i ON p.Product_ID = i.Product_ID WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);
```

PRODUCT_NAME	STOCK_QUANTITY
Bags	30

4. Average Inventory Quantity

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory FROM Inventory;
```

AVERAGE_INVENTORY
20